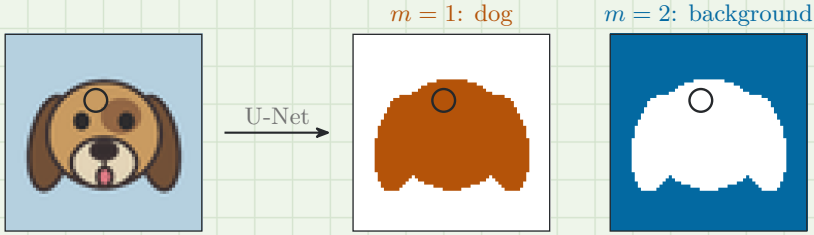
Advanced Machine Learning, ETH Zurich

U-Nets

Carlos Cotrini

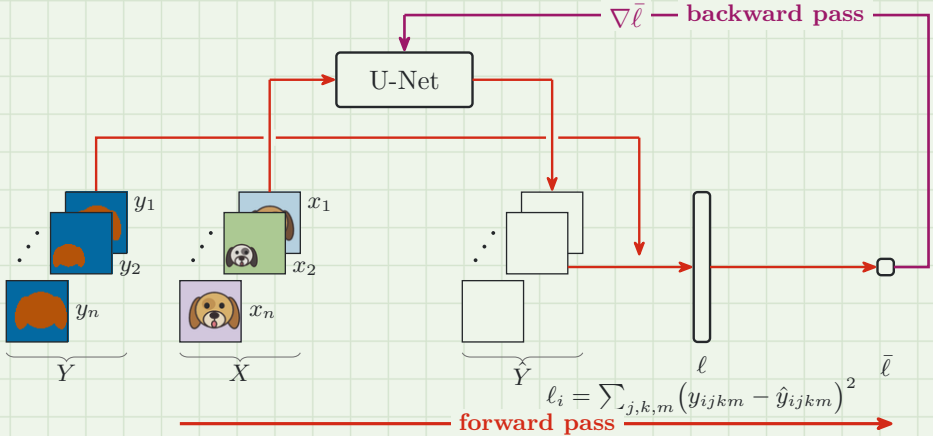
Autumn semester 2026

Image segmentation: for every pixel of an image, find the object it shows. Our example has two objects, a dog and the background.



$y_{jkm} = 1$ if pixel (j, k) shows object m, and $y_{jkm} = 0$ otherwise.

The U-Net turns each image x_i into $\hat{y}_i$, a guess of its map y_i. The loss ℓ_i adds up the squared differences over every pixel (j, k) and every channel m.



- A typical architecture for image segmentation.
- One can also treat a U-net as some sort of image autoencoder
- Image from original paper: https://arxiv.org/abs/1505.04597

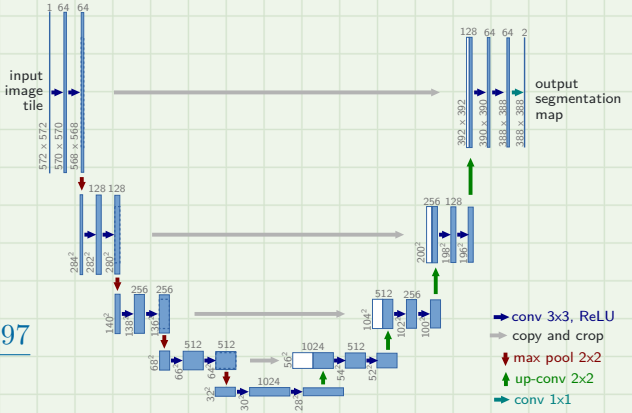
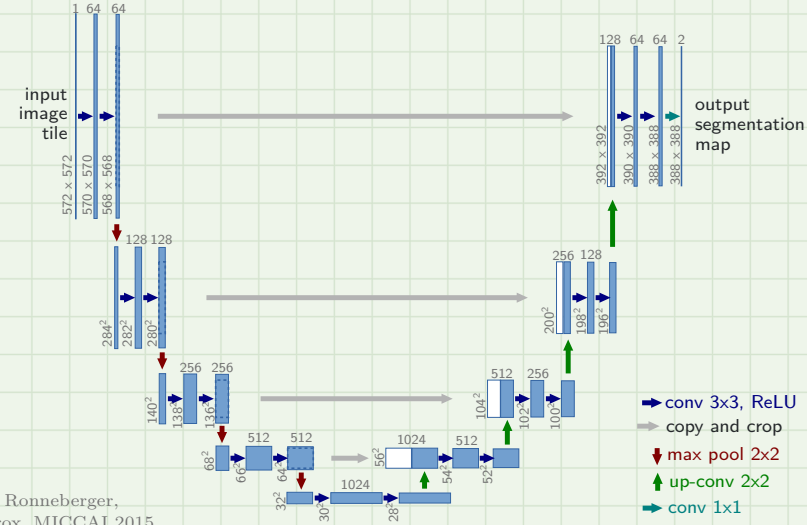


Fig. 1. U-net architecture (example for 32x32 pixels in the lowest resolution). Each blue box corresponds to a multi-channel feature map. The number of channels is denoted on top of the box. The x-y-size is provided at the lower left edge of the box. White boxes represent copied feature maps. The arrows denote the different operations.

After Fig. 1 of Ronneberger, Fischer and Brox, MICCAI 2015



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